

# Onyx — The Kits: a developer's guide

*Onyx · multi-process OS on Raspberry Pi 4*

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How an Onyx program is built on the system's shared libraries — the **kits** — with a short example for each. Each kit has its reference, with every operation it exposes (documents 10 to 19, linked in the table below). Companion of the [Developer Guide](#) (the build, the application model, the ports) and of [Kernel Internals](#).

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## 1. One kit per domain

An Onyx program does not talk to the kernel, and does not carry copies of common code. It is built on **kits**: shared libraries, one per domain, loaded once for the whole system and bound to each program that uses them.

| Kit           | Domain                                                                                                         | On the card                    | Sources                       | Reference                   |
|---------------|----------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------|-----------------------------|
| <b>AppKit</b> | What makes a program run: the system's calls, strings, the console, <code>.ini</code> files, starting programs | <code>SD:/lib/appkit.so</code> | <code>user/Kits/appkit</code> | <a href="#">10 — AppKit</a> |
| <b>UIKit</b>  | The interface: windows,                                                                                        | <code>SD:/lib/uikit.so</code>  | <code>user/Kits/uikit</code>  | <a href="#">11 — UIKit</a>  |

| Kit               | Domain                                                                                                                                                        | On the card                        | Sources                           | Reference                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------|---------------------------------|
|                   | widgets, dialogs, the theme, icons                                                                                                                            |                                    |                                   |                                 |
| <b>SystemKit</b>  | Talking to the system and the other programs: notifications, clipboard, trash, volume, wallpaper, file associations, the dock, the language and the time zone | <code>SD:/lib/systemkit.so</code>  | <code>user/Kits/systemkit</code>  | <a href="#">12 — SystemKit</a>  |
| <b>NetKit</b>     | The network: HTTP, the FTP volumes                                                                                                                            | <code>SD:/lib/netkit.so</code>     | <code>user/Kits/netkit</code>     | <a href="#">13 — NetKit</a>     |
| <b>FileKit</b>    | Files and folders, paths, compression, archives                                                                                                               | <code>SD:/lib/filekit.so</code>    | <code>user/Kits/filekit</code>    | <a href="#">14 — FileKit</a>    |
| <b>ImageKit</b>   | Pictures: reading, writing, resizing, adjusting                                                                                                               | <code>SD:/lib/imagekit.so</code>   | <code>user/Kits/imagekit</code>   | <a href="#">15 — ImageKit</a>   |
| <b>AudioKit</b>   | Sound: playing files, notes, synthesis, the sound output                                                                                                      | <code>SD:/lib/audiokit.so</code>   | <code>user/Kits/audiokit</code>   | <a href="#">16 — AudioKit</a>   |
| <b>FontKit</b>    | Fonts: FreeType, the font manager, anti-aliased text                                                                                                          | <code>SD:/lib/fontkit.so</code>    | <code>user/Kits/fontkit</code>    | <a href="#">17 — FontKit</a>    |
| <b>PrinterKit</b> | Printing: the Print dialog, a job's pages                                                                                                                     | <code>SD:/lib/printerkit.so</code> | <code>user/Kits/printerkit</code> | <a href="#">18 — PrinterKit</a> |
| <b>GPIOKit</b>    | The Raspberry Pi's 40-pin header: pins,                                                                                                                       | <code>SD:/lib/gpiokit.so</code>    | <code>user/Kits/gpiokit</code>    | <a href="#">19 — GPIOKit</a>    |

| Kit | Domain                                  | On the card | Sources | Reference |
|-----|-----------------------------------------|-------------|---------|-----------|
|     | PWM, edges, I2C, SPI; a simulated board |             |         |           |

Three rules follow from this layout, and they hold for every new development:

- **Reusable code goes into the adequate kit** — not into a new shared header, not copied into an application. If no kit fits the domain, a new kit is created (§13).
- **A program draws on the kits as much as it can**, and never reaches the kernel itself. Only AppKit reads the kernel's table; so the kernel can change without any program being rebuilt. (One exception, decided by the user: **GPIOKit** calls the kernel's `gpio_ctl` entry itself — it ships with the kernel, as AppKit does: §12.)
- **What one program alone uses stays beside that program**, in its own folder.

A kit has **one header**, `<kit>/<kit>.h`, which **declares**; the code is in the library. The memory cost of a kit is paid once: its code is shared by every program that uses it.

## 2. Using a kit in a program

### Including

`user/Kits` is on every include path. A source includes a kit's header by the kit's name, whatever its own folder:

```
#include "appkit/appkit.h"      // AppKit
#include "uikit/uikit.h"        // UIKit
#include "systemkit/systemkit.h" // SystemKit
#include "netkit/netkit.h"      // NetKit
#include "filekit/filekit.h"    // FileKit
#include "imagekit/imagekit.h"  // ImageKit
#include "audiokit/audiokit.h"  // AudioKit
#include "fontkit/uikitface.h"  // FontKit
#include "printerkit/printerkit.h" // PrinterKit
```

### Linking

AppKit needs nothing: the kernel loads it and binds it to every program. Every other kit is linked through its **import archive**, a few bytes of stubs and a constructor that opens the library before `main`:

| The program is       | It links                             |
|----------------------|--------------------------------------|
| C++ (an application) | <code>lib/&lt;kit&gt;.imp.a</code>   |
| C (a console tool)   | <code>lib/&lt;kit&gt;.imp_c.a</code> |

An archive only brings what the program calls: listing a kit a program does not use costs nothing, and the library is not opened. The applications' rules in `user/Makefile` already list UIKit, FileKit, SystemKit and NetKit; a console tool names its kits in `user/BinUtils/Makefile`:

```
# a C tool that uses SystemKit
mytool.elf: KITLIBS = ../lib/systemkit.imp_c.a
mytool.elf: ../lib/systemkit.imp_c.a
```

## Versions

A kit's table only grows: a function is never removed nor renamed, so a newer library always runs an older program. A program built against a table that grew needs at least that library: its package says so (`needs = systemkit >= 1.56` in `tools/pkg/packages.ini`), and the package manager installs the library first.

## From BASIC

A BASIC program uses the kits too: `#import filekit`, then `FileKit.copy (a$, b$, 0, 0)` — every function of a kit that takes and returns numbers, strings and pointers, by its name, and its structures as `TYPEs` (`DIM e AS FileKit.zip_entry`) (docs/04 §13, *The system's kits*). BASIC knows a kit by its **description**, `SD:/lib/<kit>.bi`, made from the kit's `.abi` and its headers by `tools/kitbi/kitbi.py` when the kits are built: nothing in BASIC names a kit, so **a new kit is importable as soon as it is built** (docs/03, *BASIC and the kits*). What BASIC cannot call — C++ classes, structures passed by value, a structure with a union or a bit field — is left out of the description: a kit meant for BASIC too exposes plain C functions.

## On a PC

The same sources build on a PC for the tests and the screenshots. There is no shared library there: the kits' headers bring their code inline, and UIKit's sources are compiled with the program. Nothing changes in the program's code.

## 3. AppKit — what makes a program run

`#include "appkit/appkit.h"` — nothing to link.

AppKit is the program's link to the system: files, processes, time, windows, sockets... (the `kapi_*` calls, listed in the Developer Guide), plus the small services every program needs: strings without a C library, the console, a reader of `.ini` files, the starting of other programs.

**A console tool:** read a file, print to the console.

```
#include "appkit/appkit.h"
```

```
int main (void)
{
    char buf[256];
    void *f = kapi_open ("SD:/etc/autostart");
    if (f == 0) { ax_putln ("no such file"); return 1; }
    int n = kapi_read (f, buf, sizeof buf - 1);
    kapi_close (f);
    buf[n > 0 ? n : 0] = '\0';

    ax_putln ("SD:/etc/autostart says:");
    ax_putln (buf);
    return 0;
}
```

**Settings from an `.ini` file** in the program's own folder:

```
if (app_ini_load ("config.ini") >= 0)
{
    const char *name = app_ini_get ("window", "title", "Untitled");
    int width       = app_ini_get_int ("window", "width", 640);
    for (int i = 0; i < app_ini_count (); i++) // or every entry, in the
file's order
```

```
    ax_putln (app_ini_key (i));
}
```

**Strings** (no C library needed):

```
char path[96]; int n = 0;
ax_strcat (path, sizeof path, &n, "SD:/docs/"); // never overflows, always
terminated
ax_strcat (path, sizeof path, &n, "notes.txt");
char num[12]; ax_itoa (42, num);
if (ax_streq (num, "42")) ax_putln (path);
```

**Starting a program** — an application by its name, or a file by whatever runs it (a `.bas` program by BASIC, a game by its emulator):

```
lx_launch ("tinypad", "SD:/docs/notes.txt"); // an app, with its arguments
lx_open ("SD:/games/tetris.gb", 0); // a file: by its runner
```

**The volumes, USB sticks** (kapi v93) — list them, eject a stick, format one:

```
struct kapi_volume v[16];
int n = kapi_vol_list (v, 16, KAPI_VOLS_ROOM); // SD:, SD1:..., USB1:...,
USB1P1:..., RAM:
for (int i = 0; i < n && i < 16; i++)
    if ((v[i].flags & KAPI_VF_REMOVABLE) && v[i].state == KAPI_VST_MOUNTED)
        ax_putln (v[i].label); // a stick plugged in: "USB" +
its label

if (kapi_vol_eject ("USB1:", 0) == -KAPI_EBUSY) // files open on it (they were
synced)
    kapi_vol_eject ("USB1:", KAPI_EJECT_FORCE); // ... after asking the user

struct kapi_format f = { KAPI_FMT_EXFAT, 0, 0, "PHOTOS" };
kapi_vol_format ("USB1:", &f); // erases it; SD: is always
refused
```

## 4. UIKit — the interface

`#include "uikit/uikit.h"` — link `lib/uikit.imp.a` (the applications' rules do).

A window is a `Root`; widgets are added to it; `run ()` is the event loop. Everything is in the namespace `uikit`.

**A window, a label, a button:**

```
#include "appkit/appkit.h"
#include "uikit/uikit.h"
using namespace uikit;

static Label *g_label;
static int g_count;

static void onClick (Widget &)
{
    char text[32]; int n = 0;
    ax_strcat (text, sizeof text, &n, "clicked ");
    char num[12]; ax_itoa (++g_count, num);
    ax_strcat (text, sizeof text, &n, num);
    g_label->setText (text); // the widget redraws itself
}
```

```
int main (void)
{
    Root root (320, 120, "Hello"); // a decorated window
    g_label = new Label (12, 12, 296, 20, "Press the button");
    root.addChild (g_label);
    root.addChild (new Button (12, 44, 100, 28, "Click me", onClick));
    root.run (); // until the window is closed
    return 0;
}
```

**Dialogs** are modal calls:

```
if (uk_messagebox ("Delete", "Delete this file?", MB_YESNO))
{
    char path[256];
    if (uk_file_open (path, sizeof path, "SD:/docs")) // the file chooser
        g_label->setText (path);
}
```

```
unsigned colour = 0x2060C0;
if (uk_color_dialog (&colour, "Pick a colour")) root.setBg (colour);
```

**Tabs** (TabStrip, uikit/tabstrip.h): a row of closable tabs over the program's own view — the strip shows them, the program shows the chosen one's content. Each tab carries a pointer for the program; a close is only **asked** (the cross, a middle click): the program decides, then removes it.

```
static TabStrip *g_tabs;
static void onTab (Widget &) { show (g_tabs->data (g_tabs->selected)); }
static void onClose (Widget &) { int i = g_tabs->closing; Doc *d = (Doc *) g_tabs->data (i);
// (a question first, if need be)
    if (doc_close (d)) g_tabs->remove (i); }
static void onNew (Widget &) { g_tabs->select (g_tabs->add ("Untitled", new_doc ()), true); }
...
g_tabs = new TabStrip (0, 0, root.width, 30, onTab);
g_tabs->onClose = onClose; g_tabs->onNew = onNew;
g_tabs->anchor = ANCHOR_LEFT | ANCHOR_TOP | ANCHOR_RIGHT;
root.addChild (g_tabs);
```

setTitle (i, s), setMark (i, on) (a dot: something runs, something new), selectNext (±1) (the program's Ctrl+Tab), activeFace (the chosen tab opens onto the content's colour). The Terminal's tabs.

**The theme:** draw with the palette (C\_BG, C\_TEXT, C\_ACCENT, C\_FACE...), never with fixed colours, so that the program follows the user's theme.

**Text in an owner-drawn widget** (uikit/text.h): uk\_text\_fit cuts a line with "...", uk\_text\_wrap word-wraps a paragraph into lines of a width (Stickies' cards), uk\_text\_over writes straight on a see-through window — the wallpaper — blended by coverage, with a soft shadow or engraved (the agenda's and Stickies' headers). The tool bar's icons (uikit/toolbar.h, WKT\_\*) include WKT\_TRASH and WKT\_PIN:

```
int st[6], ln[6]; bool more;
int k = uk_text_wrap (text, len, 196, 6, st, ln, &more); // up to 6 lines of 196 px
for (int i = 0; i < k; i++) { char l[256]; int n = ln[i] < 255 ? ln[i] : 255;
    memcpy (l, text + st[i], n); l[n] = 0; uk_text (canvas, 22, 33 + 17 * i, l, ink);
}
```

```
uk_text_over (canvas, 38, 9, "Pinned notes", 0x00FAFCFF, 2, 1); // bold, a soft shadow, on the wallpaper
```

**Without C++** (uikit/flat.h, in uikit/uikit.h): the same window and widgets as handles, behind plain C functions — what a BASIC program calls (#import UIKit), what QBStudio's generated code is made of, and what a C program can use:

```
void *win = uk_window ("Hello", 320, 120, 0);
uk_label (win, 12, 12, 200, 20, "Your name:");
void *name = uk_textbox (win, 12, 36, 200, 24, "", 0);
uk_button (win, 220, 36, 80, 24, "OK", on_ok); /* void on_ok (void *button) */
uk_window_run (win); /* until the window is closed */
```

```
#import UIKit
win = UIKit.window("Hello", 320, 120, 0)
ok = UIKit.button(win, 220, 36, 80, 24, "OK", ADDRESSOF(Clicked))
DO WHILE UIKit.window_wait(win): LOOP
```

uk\_panel (window, x, y, w, h) is an area with widgets of its own — every maker takes the window *or a panel* as what holds the widget — a page shown in place of another (uk\_show), moved elsewhere (uk\_set\_parent). QBStudio's user controls are panels.

A widget added to UIKit that BASIC should reach gets its functions there (a maker, and uk\_set\_text / uk\_get\_value ... taught its kind).

**An icon** — any picture ImageKit reads (BMP today, PNG tomorrow), given as icons are drawn (magenta = see-through):

```
#include "uikit/bmp.h"

int w, h;
unsigned *px = ui::icon_load ("SD:/apps/tinypad.app/icon.bmp", &w, &h);
if (px) { /* draw it ... */ delete [] px; }
```

**A code editor** (uikit/codeedit.h, 2026-10-06 — QBStudio's, made a widget of the kit for every program that edits code: QBStudio, Turtle Quest): BASIC in a monospaced face with its colours, the line numbers, the indentation's guides, a line lit (hiLine: an error's, a debugger's current line), the lines with a problem marked, undo / redo, the clipboard, the completion, UTF-8 comments:

```
CodeEdit *ed = new CodeEdit (10, 10, 400, 300);
ed->mono = myMonoFace; // (0: the UI's face)
ed->isKeyword = is_basic_word; // bool (const char *w, int n): the words coloured and put in capitals
ed->setText ("FOR i = 1 TO 3\n PRINT i\nNEXT\n");
ed->onChange = changed; // Action: the text changed
root.addChild (ed);
ed->hiLine = 2; ed->showLine (1); // line 2 lit and shown (hiLine 1-based, showLine 0-based)
ed->clearMarks (); ed->addMark (3); // line 3 has a problem
```

The completion: set complete (void (CodeEdit &, const char \*object)), which calls ed.addCompletion (name, kind, detail) for each item offered after object. (or Ctrl+Space: object empty).

**A timer or polling:** derive from Root and override onTick () (called about 60 times a second).

## 5. SystemKit — talking to the system and the other programs

`#include "systemkit/systemkit.h"` — link `lib/systemkit.imp.a` (C++) or `lib/systemkit.imp_c.a` (C). The one header brings every subject below (a C program gets those written in C: notifications, volume, preload list, autostart, the applets' protocol).

**A notification** (a bubble under the menu bar; `notify_action` adds what a click starts):

```
#include "systemkit/systemkit.h"

notify ("Backup", "3 files copied");
notify_action ("Updates", "2 updates available", "control pkgman");
```

**The clipboard**, shared by every application:

```
#include "systemkit/systemkit.h"

clip_set_text ("copied text");

char buf[1024];
if (clip_get_text (buf, sizeof buf)) { /* paste buf */ }

int w, h;
unsigned *px = clip_get_image (&w, &h);           // a picture, if one was copied
if (px) { /* ... */ delete [] px; }
```

**The trash** — a file is moved there instead of being deleted:

```
#include "systemkit/systemkit.h"

if (trash_move ("SD:/docs/old.txt")) notify ("Trash", "old.txt moved to the trash");
int n = trash_count ();                      // what the trash holds
```

**Opening a file with the right application** (the user's file associations):

```
#include "systemkit/systemkit.h"

char app[32];
if (fa_app_for ("SD:/photos/cat.png", app, sizeof app)) { /* "imageview" */ }
fa_open ("SD:/photos/cat.png");              // a folder, an app, a
document: opened
```

**The volume, the wallpaper, the dock** — a settings program changes the file and tells the system:

```
#include "systemkit/systemkit.h"

volume_save (7, 0);                          // 0..10, not muted: kept for
the next start

Wallpaper wp;
wp_load (wp);                                // SD:/etc/wallpaper.ini
/* change wp ... */
wp_save (wp);

dock_reload ();                              // the dock reads its settings
again
```

**Starting a program at every boot** — a line of `SD:/etc/autostart` looked for, or added where it belongs (after a given line, else before `preload /boot`, which stays last; Setup's held-back `#setup:` lines respected), and the user told so:

```
#include "systemkit/systemkit.h"
```

```
int r = autostart_ensure ("run stickies", "run agenda", "# Stickies: the pinned notes
on the desktop");
/* 2 added, 1 already there (autostart_has), 0 not written */
```

**The system's language** (the Control Panel's Language & Region): an app's words go through UIKit's TR ("`...`") after `uk_lang_init ()` (section 3); a program with words of its own asks SystemKit:

```
const char *code = locale_language (); // "en", "fr" (system.ini's
language=; none: "en")
for (int i = 0; i < locale_language_count (); i++) // the languages Onyx speaks,
each in its own words
    list->add (locale_language_name (i));
locale_set_language ("fr"); // kept: the programs started
next speak it
locale_set_zone (0); // a time zone
(locale_zone_count / _city / _offset): now, and kept
```

| Its part (a header of its own, beside systemkit.h)                   | Subject                                                                 |
|----------------------------------------------------------------------|-------------------------------------------------------------------------|
| <code>notify.h</code>                                                | Notifications                                                           |
| <code>clipboard.h</code> ( <code>clippproto.h</code> : its protocol) | The clipboard                                                           |
| <code>trash.h</code>                                                 | The trash                                                               |
| <code>fileassoc.h</code>                                             | Which application opens which file                                      |
| <code>volume.h</code>                                                | The master volume, the output, each program's volume                    |
| <code>wallpaper.h</code>                                             | The wallpaper's settings and its painter                                |
| <code>dockconf.h</code>                                              | The dock's settings                                                     |
| <code>preloadini.h</code>                                            | The programs loaded ahead at boot                                       |
| <code>autostart.h</code>                                             | The programs started at boot ( <code>SD:/etc/autostart</code> )         |
| <code>applet_proto.h</code>                                          | A settings applet shown inside the Control Panel                        |
| <code>locale.h</code>                                                | The system's language and time zone ( <code>SD:/etc/system.ini</code> ) |

## 6. NetKit — the network

`#include "netkit/netkit.h"` — link `lib/netkit.imp.a` (C++) or `lib/netkit.imp_c.a` (C). The `HttpClient` class is apart: `#include "netkit/http.hpp"`.

**A page fetched** (HTTP/1.0, no allocation: the caller gives the buffer):

```
#include "appkit/appkit.h"
#include "netkit/netkit.h"
```

```
static char buf[32 * 1024];

int main (void)
{
    http_response r;
    if (http_get ("http://example.com/", buf, sizeof buf, &r) < 0 || !r.ok)
    {
        ax_putln ("request failed");
        return 1;
    }
    kapi_stdout_write (r.body, (unsigned) r.body_len);
    return 0;
}
```

**A web API** (HTTP/1.1, headers, HTTPS with the TLS transport) — the `HttpClient` class:

```
#include "netkit/http.hpp"

static char buf[64 * 1024];
HttpClient http;
http.user_agent ("MyApp/1.0").accept ("application/json");
HttpResponse r = http.get ("http://api.example.com/status", buf, sizeof buf);
if (!r.is_error ()) { /* r.status, the body in buf */ }
```

**An FTP server as a volume:** once logged in, `FTP:host/dir/file` is a path like any other, for every program.

```
#include "netkit/netkit.h"

if (ftpfs_login ("ftp.example.com", "me", "secret", 1)) // 1: remembered
{
    void *f = kapi_open ("FTP:ftp.example.com/pub/readme.txt");
    /* kapi_read ... kapi_close */
}
```

The raw sockets (`kapi_tcp_connect`, `kapi_tcp_send...`) are AppKit's; NetKit is where protocols built on them belong.

## 7. FileKit — files, folders, archives

`#include "filekit/filekit.h"` — link `lib/filekit.imp.a`.

**A whole file, read and written:**

```
#include "filekit/filekit.h"

void *data; unsigned n;
if (fk_load ("SD:/docs/notes.txt", &data, &n) == 0)
{
    /* data: n bytes, a 0 after them */
    fk_save ("SD:/docs/notes.bak", data, n);
    fk_free (data);
}
```

**Folders and paths:**

```
#include "filekit/filekit.h"

char path[FS_PATHL];
```

```

fs_join (path, sizeof path, "SD:/docs", "report.txt");
if (!fs_exists (path)) { /* ... */ }

fk_mkdirs ("SD:/backup/2026/docs"); // every folder of the path
fk_copy ("SD:/docs", "SD:/backup/2026/docs", 0, 0); // a file, or a folder and all
it holds
fk_remove ("SD:/tmp/work"); // the same, removed

```

### A ZIP archive, read:

```

char err[80];
fk_zip *z = fk_zip_open ("SD:/downloads/pack.zip", err, sizeof err);
if (z)
{
    struct fk_zip_entry e;
    for (int i = 0; i < fk_zip_count (z); i++)
        if (fk_zip_entry (z, i, &e) ax_putln (e.name);
    fk_zip_extract_all (z, 0, "SD:/downloads/pack", 0, 0);
    fk_zip_close (z);
}

```

### A ZIP archive, written:

```

fk_zipw *w = fk_zipw_create ("SD:/backup/docs.zip");
fk_zipw_add (w, "SD:/docs", 0); // the folder and all it holds
fk_zipw_close (w, 0, 0, err, sizeof err);

```

Other archive formats (tar, tar.gz, gzip) are read through the same calls (fk\_arc\_\*); a program asks the library which formats it handles (fk\_arc\_formats) instead of keeping its own list.

**A settings or progress file with sections, read and written back** (filekit/kvtext.h, C and C++): the entries stay in the file's order, the keys a program does not know are kept, values may hold new lines.

```

fk_kv *kv = fk_kv_load ("SD:/apps/game.app/progress.ini", FK_KV_ESCAPES);
if (!kv) kv = fk_kv_new (FK_KV_ESCAPES); // no file yet: a fresh
start
int stars = atoi (fk_kv_get (kv, "Player", "level1", "0"));
fk_kv_set (kv, "Player", "level1", "3");
fk_kv_set (kv, "", "last", "level1"); // ": before the first
[header]
fk_kv_save (kv, "SD:/apps/game.app/progress.ini", "# my game's progress");
fk_kv_free (kv);

```

A level pack (FK\_KV\_PIPES): many [level] blocks (fk\_kv\_block, fk\_kv\_blocks, fk\_kv\_block\_name), a value going on over the | lines that follow, and the line of each value (fk\_kv\_line) for an error message. Circuits reads its packs and its progress.ini so.

## 8. ImageKit — pictures

```
#include "imagekit/imagekit.h" — link lib/imagekit.imp.a.
```

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written.

### A picture read, turned, saved in another format:

```

#include "imagekit/imagekit.h"

ik_image *im = ik_load ("SD:/photos/cat.jpg", 0);

```

```

if (im)
{
    int w = ik_width (im), h = ik_height (im);
    unsigned *px = ik_pixels (im);           // w * h pixels, 0xAARRGGBB
    ik_rotate (im, 1);                       // a quarter turn clockwise
    ik_save (im, "SD:/photos/cat.png", 90);  // the format: the extension's
    ik_image_free (im);
}

```

**Pixels of the program's own, saved:**

```

ik_image *shot = ik_image_from (pixels, width, height, width);
ik_save (shot, "SD:/screenshots/shot.png", 0);
ik_image_free (shot);

```

**The formats, asked** — a file chooser's filter follows what the library knows:

```

struct ik_format f[16];
int n = ik_formats (f, 16);

```

Resizing (`ik_scale`), flipping (`ik_flip`), the photo adjustments and the EXIF orientation are in the same header.

## 9. AudioKit — sound

```
#include "audiokit/audiokit.h" — link lib/audiokit.imp.a.
```

Several programs play at once: the system mixes them, each with its own volume (Control Panel ► Sound).

**A sound file** (WAV, MP3, OGG, FLAC, MIDI, FM Tracker songs), played in the background:

```

#include "audiokit/audiokit.h"

if (ak_play ("SD:/music/song.mp3", 0) < 0) ax_putln (ak_play_error ());
ak_play_volume (80);                               // 0..100
/* ... the program goes on ... */
ak_play_stop ();

```

**Notes** (the General MIDI instruments, a SoundFont):

```

ak_note_on (0, 60, 100);                           // channel, key (60 = middle
C), velocity
kapi_msleep (400);
ak_note_off (0, 60);

```

**A sound effect** (the FM synthesizer's voices — what the games use):

```

ak_fm_start (0, 440000, 0, 200);                     // voice 0, 440 Hz (in milli-
Hz), volume 0..255
kapi_msleep (120);
ak_fm_stop (0);

```

**The program's own samples** (an emulator, a synthesizer): stereo, 16 bits, 44.1 kHz (AUDIOKIT\_RATE).

```

if (ak_out_open (256, 1024) == 1)
{
    short frames[256 * 2];
    /* fill frames ... */
    ak_out_write (frames, 256);                     // waits for room
}

```

```
    ak_out_close ();
}
```

## 10. FontKit — fonts and text

#include "fontkit/uikitface.h" or "fontkit/fonts.h" — link lib/fontkit.imp.a.

FontKit is FreeType for the applications, with the card's fonts (SD:/res/fonts).

**Anti-aliased text in a UIKit application** — one call before the widgets are built:

```
#include "uikit/uikit.h"
#include "fontkit/uikitface.h"
using namespace uikit;

int main (void)
{
    ft_uikit_install ("DejaVu Sans", 13);           // every widget's text from now
on
    Root root (400, 200, "Smooth text");
    root.addChild (new Label (12, 12, 376, 24, "FreeType draws this"));
    root.run ();
    return 0;
}
```

**The font manager** — the families installed, a font at a size, text measured (a word processor, a canvas):

```
#include "fontkit/fonts.h"

fnt::init ();
for (int i = 0; i < fnt::count (); i++) ax_putln (fnt::name (i));

int fam = fnt::find ("DejaVu Serif");
fnt::Font *f = fnt::get (fam, 0, 14 * 64);           // style 0 (regular), 14 px in
26.6
int width = fnt::advance (f, 'A');                  // in 26.6 units
```

## 11. PrinterKit — printing

#include "printerkit/printerkit.h" — link lib/printerkit.imp.a.

A program prints in four steps: the Print dialog, a job, its pages, the end. Lengths are points (1/72 inch) from the page's top-left corner; colours are 0xRRGGBB. The pages are recorded: the print service turns them into what the printer takes — or into a PDF file.

```
#include "printerkit/printerkit.h"

PrintSetup s;
print_setup_default (&s);
PrintDialogInfo di = { sizeof di, "My document", 1, 0, 0, 0, 0 };
if (print_dialog (&s, &di))           // the printer, the paper, the
pages
{
    PrintJob *j = print_begin (&s, "My document");
    int font = print_font (j, "DejaVu Sans", 0);
    if (print_page (j, 0, 0))           // 0, 0: the paper chosen
    {
        print_text (j, font, 12, 72, 84, "Hello, printer", 0x000000);
    }
}
```

```

    print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
}
print_end (j); // to the print queue
}

```

Rectangles, paths (lines and curves, filled or stroked) and pictures are drawn the same way (`print_rect`, `print_path_*`, `print_image`).

## 12. GPIOKit — the 40-pin header

`#include "gpiokit/gpiokit.h"` — link `lib/gpiokit.imp.a` (C++) or `lib/gpiokit.imp_c.a` (C).

The Raspberry Pi's header from a program: a pin's mode, its level, PWM, edges, the I2C bus and SPI. Pins are **BCM GPIO numbers** (GPIO 17 is the header's pin 11: `gk_header_pin`). **The levels are 3.3 V**: never wire 5 V to a pin; at most about 16 mA a pin (an LED through a 330  $\Omega$  resistor). A pin is the program's from its first use until it gives it back (`gk_mode (pin, GK_FREE)`, `gk_release`) or ends — the system then makes it an input again, after a crash too. The system's pins (GPIO 14 / 15, the serial console; 0 / 1, the HAT EEPROM) and a pin another program has are refused: every call returns  $\geq 0$ , or a negative `GK_E*` that `gk_error` says in words.

GPIOKit is the one kit besides AppKit that calls the kernel itself (kapi v92 `gpio_ctl`; the user's exception, docs/03 §5.10); it is shipped with the kernel in the package `onyx`.

**An LED, a button (C):**

```

#include "gpiokit/gpiokit.h"

gk_mode (17, GK_OUT); // the LED: GPIO 17 -> 330 ohm
-> LED -> GND
gk_mode (27, GK_IN_PULLUP); // the button: GPIO 27 ->
button -> GND (reads 0 pressed)
gk_edges (27, GK_FALLING); // its presses queued, with
their time
for (;;)
{
    gk_event e[8];
    int n = gk_events (e, 8, 1000); // waits up to 1 s for the
first
    for (int i = 0; i < n; i++) gk_toggle (17); // each press turns the LED
over
}

```

**PWM and a servo** (GPIO 12, 13, 18, 19; 12 / 18 and 13 / 19 share a channel):

```

gk_pwm (18, 1000, 2500); // 1 kHz, 25 % high (duty in
1/10000)
gk_servo (18, 1500); // a servo: 50 Hz, a 1.5 ms
pulse (the middle)

```

**I2C** (GPIO 2 SDA, GPIO 3 SCL) — a BME280's id, a scan:

```

if (gk_i2c_open (0) == 0) // 0: 100 kHz
{
    unsigned char map[16];
    int n = gk_i2c_scan (map); // bit a of map: something
answers at a
    if (gk_i2c_reg_read (0x76, 0xD0) == 0x60) ax_putln ("a BME280 at 0x76");
    gk_i2c_close ();
}

```

**SPI 0** (GPIO 8 CE0, 7 CE1, 9 MISO, 10 MOSI, 11 SCLK): `gk_spi_open (1000000, 0)`, then `gk_spi_transfer (0, tx, rx, n)`.

**C++** has small classes that give the pin back by themselves:

```
gpiokit::Pin led (17, GK_OUT);
if (!led.ok ()) ax_putln (gk_error (led.status ()));
led.write (1);
gpiokit::I2CDevice bme (0x76);
int id = bme.reg (0xD0);
```

**No hardware?** GPIOKit has a **simulator** — a board in memory: the inputs set by `gk_sim_input`, the PWM's waves, an I2C bus with an SSD1306 display (0x3C: `gk_sim_display` gives its pixels) and a BME280 sensor (0x76), SPI looped back. `gk_sim (1)` switches to it; it runs by itself where the system has no GPIO, and on a PC. GPIO Lab (docs/04) is built on GPIOKit; BASIC has statements for it (docs/04 §13 *GPIO*).

## 13. Adding to a kit, creating a kit

**Before writing a helper, look for it in the kits.** If two programs could use it, it belongs to a kit.

### Adding a function to a kit

1. **Declare** it in the kit's header, with a comment that says what it does.
2. **Write its code** in the kit's source (for SystemKit, NetKit and `filekit/fsutil.h`: the `.inc` beside the header).
3. **Build:** the kit's table (`<kit>.abi`) gets the new name at its end. Commit the `.abi`. A name is never removed nor renamed: a function that is no longer needed keeps a body.
4. **Raise the package's version** (`tools/pkg/versions.ini`: `1.<the table's size>.0`) and what the applications need (`needs = <kit> >= 1.<size>` in `tools/pkg/packages.ini`).
5. **Document** it: its comment in the header is its documentation — run `python tools/docgen/kitdocs.py` (the kit's reference follows), add an example here if it deserves one, then `python docs/build_docs.py`.

### Creating a kit

When a domain has no kit (SystemKit and NetKit are the models — small, with no dependency):

- a folder `user/Kits/<kit>:` **one header a program includes**, `<kit>/<kit>.h` — it may bring a header per subject, each declaring its functions (`XK_API int f (...);`) with a `.inc` beside it for the code — and one source `<kit>.cpp` that compiles the code into the library;
- its rules in `user/Makefile` (copy SystemKit's block): the library `lib/<kit>.so`, its import archives `lib/<kit>.imp.a` (C++) and `lib/<kit>.imp_c.a` (C);
- its package in `tools/pkg/packages.ini` (`required = 1`, `files = lib/<kit>.so lib/<kit>.bi` — the `.bi` is what BASIC reads of it, made by the build: *From BASIC* above).

The header of a kit a C program may use is written in C.

## 14. Quick reference

| I want to...         | Kit     | Call                                                                             |
|----------------------|---------|----------------------------------------------------------------------------------|
| read or write a file | AppKit  | <code>kapi_open</code> , <code>kapi_read</code> ,<br><code>kapi_save_file</code> |
| load a whole file in | FileKit | <code>fk_load</code> , <code>fk_save</code>                                      |

| I want to...                                            | Kit                                      | Call                                                   |
|---------------------------------------------------------|------------------------------------------|--------------------------------------------------------|
| memory                                                  |                                          |                                                        |
| copy or remove a folder                                 | FileKit                                  | fk_copy, fk_remove                                     |
| build a path                                            | FileKit                                  | fs_join, fk_path_join                                  |
| read or write a ZIP                                     | FileKit                                  | fk_zip_open,<br>fk_zipw_create                         |
| read and write a settings / progress file with sections | FileKit                                  | fk_kv_load, fk_kv_set,<br>fk_kv_save                   |
| print to the console                                    | AppKit                                   | ax_puts, ax_putln                                      |
| read an .ini file                                       | AppKit                                   | app_ini_load,<br>app_ini_get                           |
| start another program                                   | AppKit                                   | lx_launch, lx_open                                     |
| open a window with widgets                              | UIKit                                    | Root, Label, Button...                                 |
| ask the user (message, file, colour)                    | UIKit                                    | uk_messagebox,<br>uk_file_open,<br>uk_color_dialog     |
| load an icon                                            | UIKit                                    | ui::icon_load                                          |
| show a notification                                     | SystemKit                                | notify, notify_action                                  |
| copy / paste                                            | SystemKit                                | clip_set_text,<br>clip_get_text                        |
| know the system's language                              | SystemKit (UIKit's TR ( ) for the words) | locale_language,<br>uk_lang_init                       |
| move a file to the trash                                | SystemKit                                | trash_move                                             |
| open a file with its application                        | SystemKit                                | fa_open                                                |
| fetch a web page                                        | NetKit                                   | http_get, HttpClient                                   |
| reach an FTP server                                     | NetKit                                   | ftpfs_login, then FTP: paths                           |
| load or save a picture                                  | ImageKit                                 | ik_load, ik_save                                       |
| resize or turn a picture                                | ImageKit                                 | ik_scale, ik_rotate                                    |
| play a sound file                                       | AudioKit                                 | ak_play                                                |
| play a note or an effect                                | AudioKit                                 | ak_note_on,<br>ak_fm_start                             |
| send samples to the output                              | AudioKit                                 | ak_out_open,<br>ak_out_write                           |
| draw smooth text                                        | FontKit                                  | ft_uikit_install,<br>fnt::get                          |
| print                                                   | PrinterKit                               | print_dialog,<br>print_begin,<br>print_page, print_end |
| light an LED, read a button                             | GPIOKit                                  | gk_mode, gk_write,<br>gk_read                          |
| wait for a button's press                               | GPIOKit                                  | gk_edges, gk_events                                    |

| I want to...                | Kit     | Call                                                      |
|-----------------------------|---------|-----------------------------------------------------------|
|                             |         |                                                           |
| dim an LED, move a servo    | GPIOKit | gk_pwm, gk_servo                                          |
| talk to an I2C / SPI device | GPIOKit | gk_i2c_reg_read,<br>gk_i2c_write_read,<br>gk_spi_transfer |